

True DGP: $y_t = \rho y_{t-1} + u_t$, $u \sim N(0, 1)$, $\rho = 0.95$, $T = 100$

Estimated OLS models:

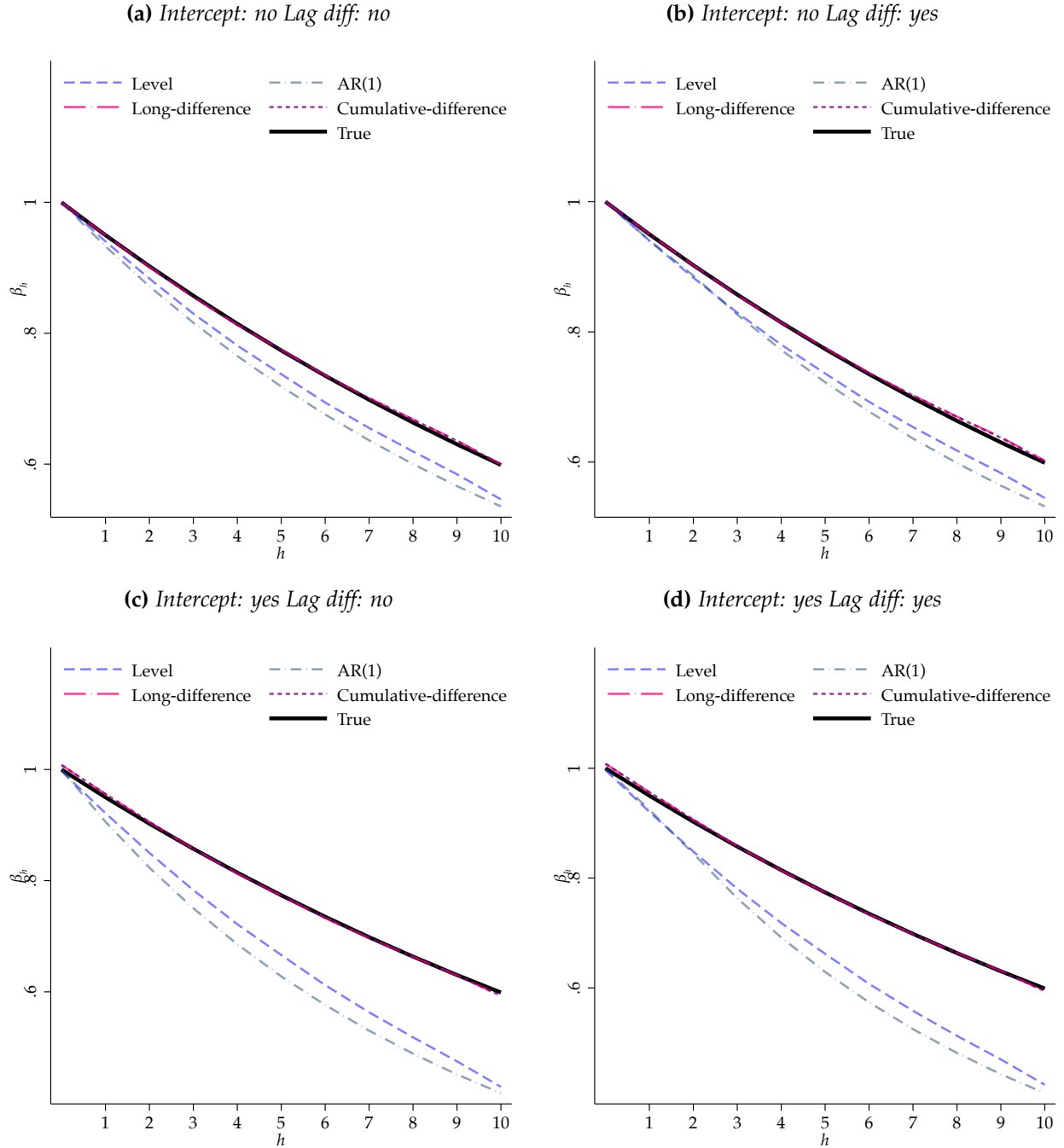
Intercept: no Lag diff: no: $y_t = \beta_1 y_{t-1} + u_t$

Intercept: no Lag diff: yes: $y_t = \beta_1 y_{t-1} + \gamma \Delta y_{t-1} + u_t$

Intercept: yes Lag diff: no: $y_t = \beta_1 y_{t-1} + \beta_0 + u_t$

Intercept: yes Lag diff: yes: $y_t = \beta_1 y_{t-1} + \beta_0 + \gamma \Delta y_{t-1} + u_t$

Figure 1: LP bias in small samples, $T = 100$, $\rho = 0.95$, $Nreps = 10,000$



True DGP: $y_t = \rho y_{t-1} + u_t$, $u \sim N(0, 1)$, $\rho = 1$, $T = 100$

Estimated OLS models:

Intercept: no Lag diff: no: $y_t = \beta_1 y_{t-1} + u_t$

Intercept: no Lag diff: yes: $y_t = \beta_1 y_{t-1} + \gamma \Delta y_{t-1} + u_t$

Intercept: yes Lag diff: no: $y_t = \beta_1 y_{t-1} + \beta_0 + u_t$

Intercept: yes Lag diff: yes: $y_t = \beta_1 y_{t-1} + \beta_0 + \gamma \Delta y_{t-1} + u_t$

Figure 2: LP bias in small samples, $T = 100$, $\rho = 1$, $Nreps = 10,000$

